Lecture 4

Covering groups, integration and descent

Objectives. *We ask when a finite-dimensional Lie algebra representation is the differential of a representation of a given connected Lie group. On a simply connected group it always is. On other connected groups we integrate on the universal covering group and then test a central kernel. For example, spin one half is a representation of* $\mathrm{SU}(2)$ *and not of* $\mathrm{SO}(3)$.

By the end of this lecture you will be able to:

- Describe the universal covering group by path classes and use the lifting theorem.
- Prove that a discrete normal subgroup of a connected group is central.
- State Lie's integration theorem and Lie III with their hypotheses, and explain path transport.
- Prove the descent criterion and apply it to $\mathbb{R} \to \mathrm{U}(1)$ and $\mathrm{SU}(2) \to \mathrm{SO}(3)$.

Lie groups are finite-dimensional, real, Hausdorff and second countable. From Lecture 1–Lecture 3 we use three facts. The map exp is a diffeomorphism near 0. A connected Lie group is generated by any neighbourhood of e. A continuous homomorphism Φ of Lie groups is smooth and satisfies $\Phi(\exp X) = \exp d\Phi(X)$. Hence two continuous homomorphisms on a connected Lie group with equal differentials are equal.

4.1 Loops, homotopy and covering groups

Definition 4.1. A *path* in a space X is a continuous map $\gamma\colon [0,1] \to X$. It is a *loop at* x if $\gamma(0) = \gamma(1) = x$. Paths γ_0, γ_1 from x to y are *homotopic* if there is a continuous map $F\colon [0,1]^2 \to X$ such that

$$F(0,t) = \gamma_0(t), \qquad F(1,t) = \gamma_1(t), \qquad F(s,0) = x, \qquad F(s,1) = y.$$

The space X is *simply connected* if it is path connected and every loop is homotopic to a constant loop.

A space is *path connected* if any two of its points are joined by a path. It is *locally path connected* if every neighbourhood of a point contains a path-connected neighbourhood of that point. Every manifold is locally path connected, and $[0,1]$ and $[0,1]^2$ are both path connected and locally path connected. A *topological group* is a group with a topology for which multiplication and inversion are continuous.

If σ starts where γ ends, the path $\gamma * \sigma$ traverses γ on $[0, \frac{1}{2}]$ and then σ on $[\frac{1}{2}, 1]$, each at double speed. Homotopy classes of loops at x form the *fundamental group* $\pi_1(X, x)$ under this product.

Definition 4.2. Let X and Y be path connected. A continuous surjection $p\colon Y \to X$ is a *covering map* if each point of X has an open neighbourhood U such that $p^{-1}(U)$ is a disjoint union of open sets, each mapped homeomorphically onto U by p.

The basic example is the map $p\colon \mathbb{R} \to \mathrm{U}(1)$ given by $p(t) = e^{it}$. Take a point $z \in \mathrm{U}(1)$ and a short open arc U around z. Then $p^{-1}(U)$ is a disjoint union of open intervals, spaced 2π apart, and p maps each interval homeomorphically onto U. So p is a covering map. Figure 4.1 draws the line as a helix, so that p becomes the vertical projection.

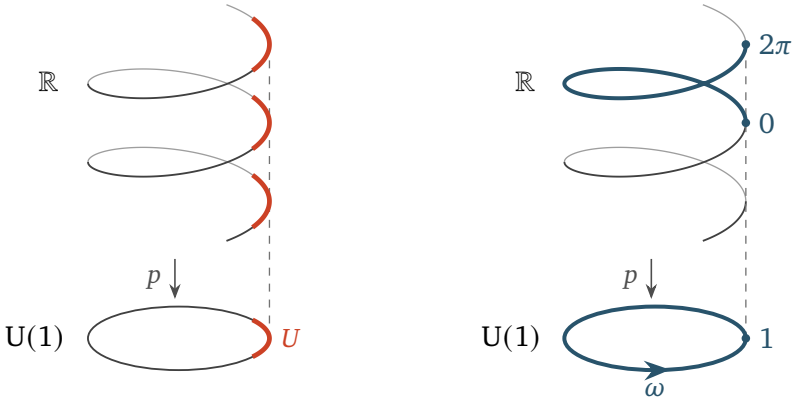


Figure 4.1: The map p projects the helix $\mathbb{R}$ vertically onto the circle $\mathrm{U}(1)$. In the first panel the coloured arcs of the helix lie over the arc U; only three of the infinitely many arcs over U are drawn. In the second panel the heavy path lies over the loop $\omega(t) = e^{2\pi i t}$ and runs from 0 to 2π, so it does not close.

Theorem 4.3 (Lifting theorem; cited). *Let $p\colon Y \to X$ be a covering map.*

(a) Let Z be path connected, locally path connected and simply connected. Let $f\colon Z \to X$ be continuous, and let $p(y_0) = f(z_0)$. Then there is a unique continuous $\tilde{f}\colon Z \to Y$ with $p \circ \tilde{f} = f$ and $\tilde{f}(z_0) = y_0$.

(b) Let γ_0, γ_1 be homotopic paths from x to x', and let $p(y_0) = x$. Then the lifts of γ_0 and γ_1 that start at y_0 have the same endpoint.

We cite this theorem from H §13.1 and S §1.2. The endpoint statement (b) follows from the lifting statement (a), applied to a homotopy F with $Z = [0,1]^2$ and $\tilde{F}(0,0) = y_0$. Each fibre of p is discrete. So if $\tilde{F}$ maps a connected set into one fibre, then $\tilde{F}$ maps that set to a single point.

The lift $\tilde{F}$ maps the edge $t = 0$ into the fibre over x, so $\tilde{F}$ maps this edge to y_0. By uniqueness of lifts in (a) with $Z = [0,1]$, the restrictions of $\tilde{F}$ to the edges $s = 0$ and $s = 1$ are the lifts of γ_0 and γ_1 from y_0.

The lift $\tilde{F}$ maps the edge $t = 1$ into the fibre $p^{-1}(x')$, so $\tilde{F}$ also maps this edge to a single point. Hence the two lifts have the same endpoint.

By the endpoint statement (b), a loop whose lift has distinct endpoints is not homotopic to a constant loop.

In the example of Figure 4.1, a path γ in $\mathrm{U}(1)$ starting at 1 lifts to a path $\tilde{\gamma}$ in $\mathbb{R}$ starting at 0. The endpoint $\tilde{\gamma}(1)$ is the *total angle* through which γ turns. The second panel shows a loop with total angle 2π.

Definition 4.4. A *universal covering group* of a connected Lie group G is a connected, simply connected Lie group $\tilde{G}$ with a smooth homomorphism $p\colon \tilde{G} \to G$ that is a covering map.

Path classes. Let G be connected. Let $\tilde{G}$ be the set of homotopy classes $[\gamma]$ of paths in G starting at e. Define

$$p[\gamma] = \gamma(1), \qquad [\gamma][\delta] = [\, t \mapsto \gamma(t)\delta(t)\,], \qquad [\gamma]^{-1} = [\, t \mapsto \gamma(t)^{-1}\,].$$

If F is a homotopy from γ to γ' and E one from δ to δ', then $(s,t) \mapsto F(s,t)E(s,t)$ is a homotopy from $\gamma\delta$ to $\gamma'\delta'$. So the product is well defined. The group axioms hold pointwise in t. The kernel of p consists of the classes of loops at e.

The pointwise product agrees with concatenation up to homotopy. Let c_x be the constant path at x. Replacing γ by $\gamma * c_{\gamma(1)}$ and δ by $c_e * \delta$ does not change their classes, since each is a reparametrisation. The pointwise product of the new paths is $\gamma * (\gamma(1)\delta)$, where $\gamma(1)\delta$ is the path $t \mapsto \gamma(1)\delta(t)$. For loops at e this is $\gamma * \delta$.

We compute this group for $G = \mathrm{U}(1)$. Two paths from 1 to the same point are homotopic exactly when they have the same total angle. If they are homotopic, the endpoint statement of the lifting theorem, Theorem 4.3(b), gives equal total angles. Conversely, if $\tilde{\gamma}_0(1) = \tilde{\gamma}_1(1)$, then

$$F(s,t) = e^{i((1-s)\tilde{\gamma}_0(t) + s\tilde{\gamma}_1(t))}$$

is a homotopy from γ_0 to γ_1, and both endpoints stay fixed. The pointwise product adds total angles, and every real number is the total angle of the path $t \mapsto e^{iat}$. So the path-class group of $\mathrm{U}(1)$ is $\mathbb{R}$ with addition. Fix θ with $e^{i\theta} = \gamma(1)$. The total angle of γ is $\theta + 2\pi k$ for an integer k. The class of γ records this integer, and the endpoint of γ does not.

It remains to give $\tilde{G}$ a topology. We sketch the construction. The details are cited after the theorem on smooth consequences, Theorem 4.5. Take a simply connected open neighbourhood U of $\gamma(1)$, for example a translate of a small exponential ball.

The basic open set $\mathcal{U}([\gamma], U)$ consists of the classes $[\gamma * \sigma]$, where σ is a path in U starting at $\gamma(1)$. The map p sends this set bijectively onto U, because paths in U with the same endpoints are homotopic in U.

The cited details show that p is a covering map, that the group operations are continuous and that $\tilde{G}$ is simply connected. Charts on G pulled back through p give the smooth structure on $\tilde{G}$.

An *isomorphism over G* from a universal covering group $p\colon \tilde{G} \to G$ to another one $p'\colon \tilde{G}' \to G$ is an isomorphism of Lie groups $\alpha\colon \tilde{G} \to \tilde{G}'$ with $p' \circ \alpha = p$.

Theorem 4.5 (Smooth consequences; cited). *Let G be a connected Lie group.*

(a) The path-class construction gives a universal covering group $p\colon \tilde{G} \to G$, unique up to isomorphism over G. Its kernel is discrete and isomorphic to $\pi_1(G, e)$, and dp is an isomorphism.

> (b) *Let G' be a connected Lie group and $q\colon G' \to G$ a surjective smooth homomorphism with dq bijective. Then q is a covering map and* $\ker q$ *is discrete.*

We cite this from H §5.8 and from Knapp Beyond, Section I.11, *Covering groups*. Hall assumes that G is a matrix group, but $\tilde{G}$ need not be one.

> **Proposition 4.6.** *Let G be a connected topological group. Let N be a normal subgroup of G that is discrete in the subspace topology. Then N lies in the centre of G.*

Proof. Fix $n \in N$ and put $c(g) = gng^{-1}$. The map $c\colon G \to G$ is continuous, because multiplication and inversion are continuous. Normality gives $c(G) \subseteq N$. Since G is connected, $c(G)$ is connected. It contains $c(e) = n$.

Since N is discrete, every point of N is open in N. In particular, $\{n\}$ is open in N. Its complement $N \setminus \{n\}$ is a union of open singletons, so $\{n\}$ is also closed in N. Hence $\{n\}$ is open and closed in the subset $c(G)$ of N. Since $c(G)$ is connected and contains n, $c(G) = \{n\}$. In other words, $gng^{-1} = n$ for every $g \in G$. ∎

By the theorem on smooth consequences, Theorem 4.5(a), $\ker p$ is a discrete normal subgroup of the connected group $\tilde{G}$. By the proposition on discrete normal subgroups, Proposition 4.6, $\ker p$ is central. Hence $\pi_1(G, e)$, which is isomorphic to $\ker p$, is abelian. Exercise 4.1 shows that connectedness cannot be dropped.

4.2 Lie's integration theorem and Lie III

> **Theorem 4.7** (Lie's integration theorem; cited, with a sketch)**.** *Let G and H be Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$. Assume that G is connected and simply connected. Let $\varphi\colon \mathfrak{g} \to \mathfrak{h}$ be a homomorphism of real Lie algebras. Then there is a unique smooth homomorphism $\Phi\colon G \to H$ with $d\Phi = \varphi$. It satisfies $\Phi(\exp X) = \exp \varphi(X)$.*

We cite this from H §5.7 for matrix groups. For general Lie groups we cite Knapp Beyond, Sections I.10–11, *Elementary theory of Lie groups* and *Covering groups*. For representations take $H = \mathrm{GL}(V)$ with $\dim_{\mathbb{C}} V < \infty$.

Path transport. Near e the map we want is given by exp. To reach a far element g, we take a path in small steps and multiply the images of the steps in order. Two paths could give different answers, and simple connectivity excludes this. The existence proof has three steps and a conclusion. Only the conclusion uses simple connectivity.

Step 1: a local homomorphism. Choose a neighbourhood B of 0 on which exp is a diffeomorphism. Define $f(\exp X) = \exp \varphi(X)$ for $X \in B$. For small X, Y we have $\exp X \exp Y = \exp Z(X, Y)$. The Baker–Campbell–Hausdorff series is the series of iterated brackets

$$Z(X, Y) = X + Y + \frac{1}{2}[X, Y] + \cdots .$$

This is the series $H(X,Y)$ of Lecture 2. We write Z here because H is the target group. Since φ preserves brackets, it sends $Z(X,Y)$ to $Z(\varphi X, \varphi Y)$. Hence, for some neighbourhood W of e,

$$f(xy) = f(x)f(y) \qquad (x, y \in W). \tag{4.1}$$

Hall proves the series for matrix groups in H §§5.3–5.6. In general we cite Knapp Beyond, Appendix B.4, Theorem B.22, *Campbell–Baker–Hausdorff formula*.

Step 2: transport along a path. Let γ start at e. Choose $0 = t_0 < \cdots < t_n = 1$ with $\gamma(s)^{-1}\gamma(t) \in W$ whenever s, t lie in one interval $[t_{j-1}, t_j]$. Define

$$\Phi_\gamma = f\big(\gamma(t_0)^{-1}\gamma(t_1)\big) \cdots f\big(\gamma(t_{n-1})^{-1}\gamma(t_n)\big).$$

Inserting a point s into $[t_{j-1}, t_j]$ replaces one factor $f(ab)$ by $f(a)f(b)$, where $a = \gamma(t_{j-1})^{-1}\gamma(s)$ and $b = \gamma(s)^{-1}\gamma(t_j)$. By the local homomorphism property (4.1), applied to a and b, the product is unchanged. Any two such partitions have a common refinement. Hence Φ_γ depends only on γ, not on the partition.

Step 3: homotopy invariance. We sketch the topological step. Let F be a homotopy from γ to γ', both from e to g. Cut $[0,1]^2$ into squares so small that $F(q)^{-1}F(q') \in W$ for any two points q, q' of one square. Give an edge from q to q' the factor $f(F(q)^{-1}F(q'))$. The transport of an edge path is the product of its factors in order. Let one square have corners a, b, c, d in cyclic order, with images x_a, x_b, x_c, x_d. By the local homomorphism property (4.1),

$$f(x_a^{-1}x_b)\, f(x_b^{-1}x_c) = f(x_a^{-1}x_c) = f(x_a^{-1}x_d)\, f(x_d^{-1}x_c).$$

So the two routes $a \to b \to c$ and $a \to d \to c$ around one square have equal transport. Figure 4.2 shows such a move.

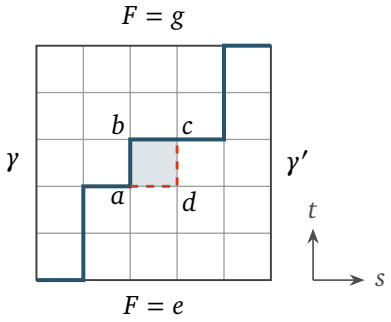


Figure 4.2: A homotopy F on a grid. The heavy edge path runs from $(0,0)$ to $(1,1)$. One move replaces its route $a \to b \to c$ by the dashed route $a \to d \to c$ across the shaded square.

Each edge path we use runs in the grid from $(0,0)$ to $(1,1)$. Start with the column $s = 0$ followed by the edge $t = 1$. End with the edge $t = 0$ followed by the column $s = 1$. Pass from the first to the last one square at a time. Each move swaps the routes $a \to b \to c$ and $a \to d \to c$ around one square, so it keeps the transport. The edges at $t = 0$ and $t = 1$ contribute $f(e) = e$, because F is constant there. By Step 2 the two columns give Φ_γ and $\Phi_{\gamma'}$. Hence $\Phi_\gamma = \Phi_{\gamma'}$. This step holds in every Lie group.

Conclusion. As G is simply connected, any two paths from e to g are homotopic. So $\Phi(g) = \Phi_\gamma$ does not depend on the path γ. The *increments* of a path δ are the elements $\delta(s)^{-1}\delta(t)$ that

enter the factors of Step 2. They do not change when δ is replaced by its left translate $g\delta$, given by $t \mapsto g\delta(t)$. So if δ runs from e to h, then

$$\Phi(gh) = \Phi_{\gamma * g\delta} = \Phi_\gamma \, \Phi_\delta = \Phi(g)\,\Phi(h).$$

Extending a path from e to g by a single short step gives $\Phi(g \exp X) = \Phi(g) \exp \varphi(X)$ for small X. Hence Φ is smooth and $d\Phi = \varphi$.

Without simple connectivity, Steps 1–3 still hold, but transport may depend on the homotopy class of the path. Self-check 2 gives an example.

Theorem 4.8 (Lie III; cited)**.** *Every finite-dimensional real Lie algebra is the Lie algebra of a connected, simply connected Lie group. This group is unique up to isomorphism.*

We cite existence from H §5.10. There Ado's theorem and H §5.9 give a connected Lie subgroup of a general linear group with the given Lie algebra. This subgroup need not be closed. The universal covering group of this subgroup is the required group. Uniqueness follows by integrating an isomorphism and its inverse with Lie's integration theorem, Theorem 4.7. The universal covering group of $\mathrm{SL}(2,\mathbb{R})$ is not isomorphic to any matrix Lie group (H §4.8). Hence Lie III needs abstract Lie groups.

4.3 Descent through a covering kernel

Theorem 4.9 (Descent)**.** *Let $p\colon \tilde{G} \to G$ be a surjective smooth homomorphism of Lie groups with dp bijective, and let $\Gamma = \ker p$. Let H be a topological group and $\tilde{\Phi}\colon \tilde{G} \to H$ a continuous homomorphism.*

(a) A homomorphism $\Phi\colon G \to H$ with $\Phi \circ p = \tilde{\Phi}$ exists if and only if $\Gamma \subseteq \ker \tilde{\Phi}$.

(b) Such Φ is unique and continuous. If H is a Lie group and $\tilde{\Phi}$ is smooth, then Φ is smooth and $d\Phi = d\tilde{\Phi} \circ (dp)^{-1}$.

Proof. Local sections. Since p is a homomorphism, left translations satisfy $p \circ L_x = L_{p(x)} \circ p$. Differentiating at e gives

$$dp_x \circ (dL_x)_e = (dL_{p(x)})_e \circ dp.$$

Both translations have invertible derivatives, and dp is bijective. Hence dp_x is bijective at every $x \in \tilde{G}$. By the inverse function theorem, each x has an open neighbourhood V_x that p maps diffeomorphically onto an open set. Let $s_x\colon p(V_x) \to V_x$ be the inverse of this restriction of p.

Necessity. If $\Phi \circ p = \tilde{\Phi}$ and $\gamma \in \Gamma$, then $\tilde{\Phi}(\gamma) = \Phi(e) = e$.

Sufficiency. Assume $\Gamma \subseteq \ker \tilde{\Phi}$. Let $g \in G$. Since p is surjective, we can write $g = p(x)$. Set $\Phi(g) = \tilde{\Phi}(x)$. If also $g = p(y)$, then $x^{-1}y \in \Gamma$ and

$$\tilde{\Phi}(y) = \tilde{\Phi}(x)\,\tilde{\Phi}(x^{-1}y) = \tilde{\Phi}(x).$$

Hence Φ is well defined and $\Phi \circ p = \tilde{\Phi}$. Moreover, Φ is a homomorphism, because

$$\Phi\big(p(x)p(y)\big) = \Phi\big(p(xy)\big) = \tilde{\Phi}(xy) = \tilde{\Phi}(x)\tilde{\Phi}(y).$$

Regularity. On $p(V_x)$ we have $\Phi = \Phi \circ p \circ s_x = \tilde{\Phi} \circ s_x$. This is continuous, and smooth when $\tilde{\Phi}$ is smooth. The open sets $p(V_x)$ cover G. Differentiating $\Phi \circ p = \tilde{\Phi}$ at e gives $d\Phi \circ dp = d\tilde{\Phi}$. Uniqueness holds because p is surjective. ∎

The proof uses only that H is a topological group. For example, the unitary group of a Hilbert space is a topological group for pointwise norm convergence. Lecture 19 uses this case.

Corollary 4.10. *Let G be a connected Lie group with universal covering group $p\colon \tilde{G} \to G$, and let $\Gamma = \ker p$. Let $\varphi\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a representation with $\dim_{\mathbb{C}} V < \infty$. Let $\tilde{\Pi}$ be the representation of $\tilde{G}$ with $d\tilde{\Pi} = \varphi \circ dp$. Then $\varphi = d\Pi$ for a representation Π of G if and only if $\tilde{\Pi}(\Gamma) = \{I\}$. In that case Π is unique and $\Pi \circ p = \tilde{\Pi}$.*

Proof. Since $\tilde{G}$ is simply connected, Lie's integration theorem, Theorem 4.7, gives $\tilde{\Pi}$, and $\tilde{\Pi}$ is unique. If Π exists, then $\Pi \circ p$ has differential $\varphi \circ dp$. Hence, by the uniqueness of $\tilde{\Pi}$, $\Pi \circ p = \tilde{\Pi}$, and $\tilde{\Pi}(\gamma) = \Pi(e) = I$ for $\gamma \in \Gamma$. Conversely, if $\tilde{\Pi}(\Gamma) = \{I\}$, then the Descent Theorem, Theorem 4.9, gives Π with $d\Pi = \varphi \circ dp \circ (dp)^{-1} = \varphi$. Uniqueness holds because G is connected. ∎

For disconnected groups the differential need not determine the homomorphism. For example, Exercise 4.6 gives two characters of $\mathrm{O}(2)$ with the same differential.

4.4 The line over the circle

Let the surjection $p\colon \mathbb{R} \to \mathrm{U}(1)$ be given by $p(t) = e^{it}$. Here $\operatorname{Lie} \mathrm{U}(1) = i\mathbb{R}$ and $\exp(i\theta) = e^{i\theta}$. Then

$$p(s+t) = p(s)\,p(t), \qquad dp(s) = \frac{d}{du}\Big|_{u=0} e^{ius} = is, \qquad \Gamma = \ker p = 2\pi\mathbb{Z}.$$

The line is simply connected, because $F(s,t) = (1-s)\gamma(t) + s\gamma(0)$ contracts any loop γ. By the theorem on smooth consequences, Theorem 4.5(b), p is a covering map, so it is a universal covering group. By the uniqueness and kernel statements of Theorem 4.5(a), $\pi_1(\mathrm{U}(1), 1) \cong 2\pi\mathbb{Z} \cong \mathbb{Z}$.

Concretely, the loop $\omega(t) = e^{2\pi i t}$ lifts from 0 to $t \mapsto 2\pi t$, which ends at $2\pi \neq 0$. By the endpoint statement of the lifting theorem, Theorem 4.3(b), ω is not homotopic to a constant loop.

A *scalar infinitesimal weight* on $\mathbb{R}$ is a number $\lambda \in \mathbb{C}$, viewed as the representation $\varphi_\lambda(s) = \lambda s$ on $\mathbb{C}$. It integrates to $\tilde{\chi}_\lambda(t) = e^{\lambda t}$.

The *period* of λ is $e^{2\pi\lambda} = \tilde{\chi}_\lambda(2\pi)$, the value at the generator 2π of Γ.

Say that λ *descends* if $\varphi_\lambda \circ (dp)^{-1}$ is the differential of a character of $\mathrm{U}(1)$. A *character* of $\mathrm{U}(1)$ here is a continuous homomorphism to the nonzero complex numbers. It is a one-dimensional representation, and its value is its trace. The descent criterion, Corollary 4.10, gives

$$\lambda \text{ descends} \iff e^{2\pi\lambda} = 1 \iff \lambda \in i\mathbb{Z}.$$

For $\lambda = in$ the descended character is $\chi(z) = z^n$, because $\chi(e^{it}) = e^{int}$. Its differential is $d\chi(i\theta) = in\theta$. In the course convention $d\chi = i\ell$ with $\ell(i\theta) = n\theta$. On $\Lambda = \ker\exp = 2\pi i\mathbb{Z}$ we get $\ell(2\pi ik) = 2\pi nk$, so $\ell(\Lambda) \subseteq 2\pi\mathbb{Z}$.

Success. For $\lambda = 3i$ the period is $e^{6\pi i} = 1$. The character $\chi(z) = z^3$ satisfies $\chi(p(t)) = e^{3it} = \tilde{\chi}_{3i}(t)$.

Failure. For $\lambda = i/2$ the period is $e^{i\pi} = -1$. Suppose a character χ had $d\chi(i\theta) = i\theta/2$. The corollary gives $\chi \circ p = \tilde{\chi}_{i/2}$, so

$$1 = \chi(1) = \chi\big(p(2\pi)\big) = \tilde{\chi}_{i/2}(2\pi) = -1.$$

This contradiction shows that $\lambda = i/2$ does not descend.

4.5 SU(2) over SO(3)

Let $\sigma_1, \sigma_2, \sigma_3$ be the *Pauli matrices*

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Put $X_j = -\frac{i}{2}\sigma_j$ for $j = 1, 2, 3$. Then $[X_1, X_2] = X_3$, with cyclic permutations.

Each element of SU(2) has the form

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

So SU(2) is homeomorphic to the unit sphere in $\mathbb{C}^2$, which is connected and simply connected (H §13.2).

The covering. The basis (X_1, X_2, X_3) is orthonormal for the form $\langle X, Y\rangle = -2\,\mathrm{tr}(XY)$. Each $\mathrm{Ad}(g)$, given by $\mathrm{Ad}(g)X = gXg^{-1}$, is a linear map of $\mathfrak{su}(2)$ to itself. It preserves this form, because the trace is invariant under conjugation. Hence the matrix of $\mathrm{Ad}(g)$ in the basis (X_1, X_2, X_3) lies in O(3). We identify $\mathrm{Ad}(g)$ with this matrix, so Ad maps SU(2) into O(3). Its image lies in SO(3), because $\det \circ \mathrm{Ad}$ is continuous on the connected group SU(2), takes values ± 1, and equals 1 at I.

The operator $\mathrm{ad}\, X_3$ of Lecture 2 sends $X_1 \mapsto X_2$, $X_2 \mapsto -X_1$ and $X_3 \mapsto 0$. Hence

$$\mathrm{Ad}(\exp\theta X_3) = e^{\theta\,\mathrm{ad}\,X_3} = R_3(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The differential $d\,\mathrm{Ad} = \mathrm{ad}$ (Lecture 2) is injective, since $[X_1, X] = [X_2, X] = 0$ forces $X = 0$. By dimension it is bijective onto $\mathfrak{so}(3)$.

Lecture 3 let SU(2) act on the traceless Hermitian matrices V by $\xi \mapsto U\xi U^*$. This action is Ad. Indeed, $\xi(x, y, z) = x\sigma_1 + y\sigma_2 + z\sigma_3$, so $-\frac{i}{2}\xi(x, y, z) = xX_1 + yX_2 + zX_3$. This linear isomorphism from V to $\mathfrak{su}(2)$ commutes with conjugation by U. The basis of Lecture 1 is

$A = -2X_3$, $B = -2X_2$ and $C = -2X_1$. The factor $-\frac{1}{2}$ gives $[X_1, X_2] = X_3$ with no factor two. Hence $k_\theta = \exp(\theta A) = \exp(-2\theta X_3)$, and

$$\operatorname{Ad}(k_\theta) = R_3(-2\theta).$$

This is the rotation by -2θ found in Lecture 3. The matrix g above is the U of Lecture 1, with $a = \alpha$ and $b = -\bar{\beta}$. Exercise 1.3 chose a different isomorphism onto $\mathfrak{so}(3)$. It differs from this one by conjugation with an element of $\mathrm{SO}(3)$.

If $\operatorname{Ad}(g) = I$, then g commutes with σ_3 and σ_1. The first condition makes g diagonal. The second makes it scalar. So $g = zI$ with $z^2 = \det g = 1$, and $\ker \operatorname{Ad} = \{\pm I\}$.

The image of Ad contains exp of a neighbourhood of 0 in $\mathfrak{so}(3)$, because $d\operatorname{Ad}$ is bijective. Hence the image of Ad is an open subgroup of $\mathrm{SO}(3)$. It is also closed, because its complement is a union of open cosets.

Every rotation has the form $kR_3(\theta)k^{-1}$ with $k \in \mathrm{SO}(3)$. Indeed, let $g \in \mathrm{SO}(3)$. Then $\det(g - I) = \det g \det(I - g^{\mathsf{T}}) = \det(I - g) = -\det(g - I)$, so $\det(g - I) = 0$ and g fixes a unit vector u. By transitivity on S^2 (Lecture 3), there is $k \in \mathrm{SO}(3)$ with $ke_3 = u$. Then $k^{-1}gk$ fixes e_3, so it is some $R_3(\theta)$ by the stabiliser computation of Lecture 3. The path $t \mapsto kR_3(t\theta)k^{-1}$ joins this rotation to I. Hence $\mathrm{SO}(3)$ is connected. The image of Ad is open, closed and nonempty, so Ad is onto.

Let $\Gamma = \ker \operatorname{Ad}$. By the theorem on smooth consequences, Theorem 4.5(b), Ad is a covering map. Since $\mathrm{SU}(2)$ is connected and simply connected, Ad is a universal covering group. By the uniqueness and kernel statements of Theorem 4.5(a), $\pi_1(\mathrm{SO}(3), I) \cong \Gamma = \{\pm I\}$.

The loop $t \mapsto R_3(2\pi t)$ lifts to $t \mapsto \exp(2\pi t X_3) = \operatorname{diag}(e^{-i\pi t}, e^{i\pi t})$, which ends at $-I$. By the endpoint statement of the lifting theorem, Theorem 4.3(b), this loop is not homotopic to a constant loop.

Figure 4.3 shows this loop. The axis-angle form draws $\mathrm{SO}(3)$ as a ball. The point θn, with n a unit vector and $0 \le \theta \le \pi$, stands for the rotation by θ about n. Opposite points of the boundary are one rotation, since a half turn about n equals a half turn about $-n$.

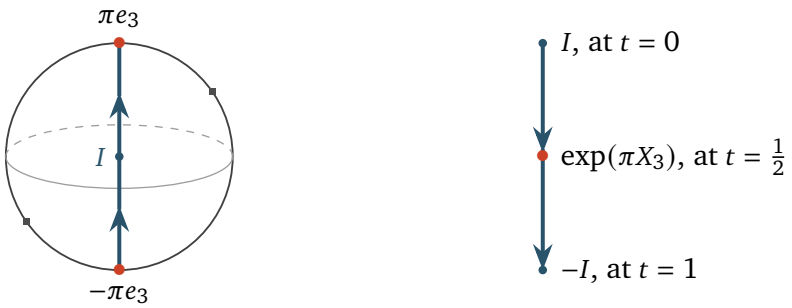


Figure 4.3: On the left, $\mathrm{SO}(3)$ as the ball of radius π. The coloured points are one rotation, and so are the square points. The heavy path is the loop $R_3(2\pi t)$: it rises from the centre to πe_3, reappears at $-\pi e_3$ and returns to the centre. On the right, the lift $t \mapsto \exp(2\pi t X_3)$ of this loop to $\mathrm{SU}(2)$ runs from I to $-I$, so it does not close.

Representations. Let V_m be the irreducible $\mathfrak{sl}(2,\mathbb{C})$-module of dimension $m+1$ and spin $m/2$. It has a basis $v_0,\dots,v_m$ with $\sigma_3 v_k = (m-2k)v_k$. Restriction to $\mathfrak{su}(2)$ gives φ_m, which integrates to Π_m on SU(2). Since $X_3 = -\frac{i}{2}\sigma_3$,

$$\Pi_m(\exp tX_3)v_k = e^{\lambda_k t}v_k, \qquad \lambda_k = -\tfrac{i}{2}(m-2k), \qquad \Pi_m(-I)v_k = e^{2\pi\lambda_k}v_k = (-1)^m v_k.$$

By the descent criterion, Corollary 4.10, $\varphi_m \circ (d\,\mathrm{Ad})^{-1}$ is the differential of a representation of SO(3) exactly when m is even. The homomorphism $t \mapsto R_3(t) = \mathrm{Ad}(\exp tX_3)$ from $\mathbb{R}$ has kernel $2\pi\mathbb{Z}$, and $\exp(2\pi X_3) = -I$. So the test is the scalar test of § 4.4 for each weight λ_k: its period $e^{2\pi\lambda_k}$ must equal 1.

Success: $m = 2$. Here $\Pi_2(-I) = I$. Hence there is a representation Π of SO(3) with $\Pi_2 = \Pi\circ\mathrm{Ad}$. This Π is equivalent to the defining representation. To see this, note that the matrices $d\,\mathrm{Ad}(X)$ also describe $\mathrm{ad}\,X$ on $\mathfrak{sl}(2,\mathbb{C})$ in the complex basis (X_1, X_2, X_3). This adjoint module has σ_3-weights $2, 0, -2$. Hence there is an invertible $E\colon \mathbb{C}^3 \to V_2$ with

$$E\,d\,\mathrm{Ad}(X)E^{-1} = \varphi_2(X).$$

The representation $g \mapsto E\,\mathrm{Ad}(g)E^{-1}$ of SU(2) has differential φ_2. By uniqueness on the connected group SU(2), it equals Π_2. Since Ad is onto, $\Pi(R) = ERE^{-1}$.

Failure: $m = 1$. Here $V_1 = \mathbb{C}^2$ and $\Pi_1(g) = g$. Suppose a representation Π of SO(3) had $d\Pi = \varphi_1 \circ (d\,\mathrm{Ad})^{-1}$. The corollary gives $\Pi \circ \mathrm{Ad} = \Pi_1$, so

$$I = \Pi\big(R_3(2\pi)\big) = \Pi\big(\mathrm{Ad}(-I)\big) = \Pi_1(-I) = -I.$$

Hence no representation of SO(3) has differential $\varphi_1 \circ (d\,\mathrm{Ad})^{-1}$.

For a spin-$\frac{1}{2}$ particle, $\mathbb{C}^2$ describes the internal spin degree of freedom. Rotations act on this space only up to sign. Lecture 13 treats such projective representations.

Self-checks

Check 1. Does the character $t \mapsto e^{i\sqrt{2}t}$ of $\mathbb{R}$ descend through $p(t) = e^{it}$ to U(1)?

Answer. No. Its period is $e^{2\pi i\sqrt{2}} \neq 1$, because $\sqrt{2} \notin \mathbb{Z}$.

Check 2. Take $G = \mathrm{U}(1)$ and $\varphi(i\theta) = i\theta/2$, so $f(e^{i\theta}) = e^{i\theta/2}$ for $|\theta| < \pi$. Use $W = \{e^{i\theta} : |\theta| < \pi/2\}$. Compute the transport around $\omega(t) = e^{2\pi i t}$. Which part of the path-transport argument fails?

Answer. The partition $t_j = j/5$ is admissible, since $2\pi/5 < \pi/2$. So $\Phi_\omega = f(e^{2\pi i/5})^5 = (e^{\pi i/5})^5 = -1$. The constant loop gives 1. Steps 1–3 remain valid. The Conclusion fails, because ω is not homotopic to the constant loop, by the lift computed in § 4.4. So $\Phi(1)$ is not well defined.

Check 3. Is φ_3 the differential of a representation of SU(2)? Is $\varphi_3 \circ (d\,\mathrm{Ad})^{-1}$ the differential of a representation of SO(3)?

Answer. Yes for SU(2), by Lie's integration theorem, Theorem 4.7. No for SO(3), since $\Pi_3(-I) = -I$. Equivalently, the weight $\lambda_0 = -\frac{3i}{2}$ has period $e^{-3\pi i} = -1$.

Summary

A connected Lie group G has a universal covering group $p\colon \tilde{G} \to G$. Its kernel $\Gamma \cong \pi_1(G, e)$ is discrete and central. Let φ be a representation of the Lie algebra of G on a finite-dimensional complex vector space. Then $\varphi \circ dp$ integrates to a representation $\tilde{\Pi}$ of $\tilde{G}$. The representation φ is the differential of a representation of G exactly when Γ acts trivially through $\tilde{\Pi}$. For $\mathbb{R} \to \mathrm{U}(1)$ this is the period condition $\lambda \in i\mathbb{Z}$. For $\mathrm{SU}(2) \to \mathrm{SO}(3)$ the condition holds exactly for even m, that is, for integer spin.

Reading

H is Hall, *Lie Groups, Lie Algebras, and Representations*, second edition, 2015. S is Sepanski, *Compact Lie Groups*, 2007. Knapp Beyond is Knapp, *Lie Groups Beyond an Introduction*, digital second edition, 2023.

- H §§4.7–4.8 and 5.7–5.10: group versus Lie algebra homomorphisms, universal covers, subgroups and subalgebras, Lie's third theorem, and a nonmatrix Lie group.
- H §§13.1–13.2: the fundamental group and the fundamental groups of the compact classical groups.
- S §§1.2, 4.2: topological background and the related material for compact groups.
- Knapp Beyond, Sections I.10–11, *Elementary theory of Lie groups* and *Covering groups*; Appendix B.4, Theorem B.22, *Campbell–Baker–Hausdorff formula*: general versions of the cited results.